

Vinay Reddy Veramareddy

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SUMMARY

Data professional with over 3 years of expertise in engineering pipelines, policy-relevant analytics, and software development. Skilled in **Python**, **SQL**, **AWS**, and **Snowflake**, with extensive experience in public sector resource management and translating complex datasets into actionable insights for legislative and technical stakeholders.

EXPERIENCE

Data Research Analyst, State of Montana, Helena, United States Jan 2024 — Feb 2025

- Produced **forecasting models** and **LLM** assisted trend analysis for **1.13M** members to enhance **Medicare** and **Medicaid** resource management using **SAS** and **Snowflake**.
- Established automated workflow orchestration by implementing **Airflow** within **AWS** environments, decreasing pipeline latency by **15%** building dashboard using **Power BI**.
- Constructed **ETL** pipelines and **statistical models** to uncover actionable insights, integrating **SQL** and **ArcGIS** for advanced **geospatial analysis**.
- Strengthened organizational alignment by consolidating **KPIs** into **Tableau** dashboards, improving data strategy.

Data Engineer, Volant Soft, Dallas, United States Mar 2022 — Jan 2024

- Developed large-scale **ML pipelines** and data ingestion processes using **Java**, **Python**, **PyTorch**, and **Apache Spark** to generate automated insights.
- Built a comprehensive **Data Lake** using the **AWS Big Data** ecosystem and **Hadoop** frameworks.
- Implemented high-throughput ingestion processes and service architectures using **Java** with **MySQL** and **PostgreSQL** to support **scalable** workloads.
- Built **Power BI** dashboards and **GCP** data migration rules to streamline reporting and ensure high data integrity.

Graduate Research Assistant, TAMUC, Commerce, United States Aug 2020 — Sep 2021

- Accelerated signal processing by **45%** by implementing **Capon** and **ROOT MUSIC** algorithms in **MATLAB**.
- Reduced computational load by **36%** through the application of novel mathematical concepts to real-time data.
- Collaborated with cross-functional teams across research initiatives to coordinate efforts, share insights, and support successful project outcomes.

Software Engineer (Intern), TLC, Hyderabad, India Jan 2019 — Mar 2019

- Streamlined supply chain reporting by **50%** through automation using **Java**, **MySQL**, and **Power BI**.
- Contributed to the design and implementation of a Data Lake using the **AWS Big Data** ecosystem and Hadoop frameworks and **SQL** to process **BigQuery** datasets, enhancing data retrieval speeds by **20%** from
- Reduced computational load by **36%** through the application of novel mathematical concepts to real-time data.
- Operated **Cassandra NoSQL** databases to manage large-scale data storage and retrieval using **Tensorflow**.

EDUCATION

University Of Cumberlands, United States - PhD in IT, Data Science - (GPA: 4, Discontinued) Jan 2025 — May 2026

TAMUC, Commerce, United States - Master of Science, Computer Science - (GPA: 3.79) Jan 2020 — Dec 2021

INDU, Hyderabad, India - Bachelor of Engineering , ECE - (GPA: 3.69) Jun 2015 — Jun 2019

SKILLS

Programming Languages : SQL, Python, Java, Apache Spark, SAS, C++, R

Cloud and Big Data Technologies : AWS (S3, EC2, Athena, Crawler), GCP, HDFS, Hive, Airflow, PySpark, Kafka, Docker, Kubernetes

Databases/Tools : PostgreSQL, MySQL, HANA, Snowflake, MongoDB, Cassandra, Git

AI/Analysis : Agentic AI, Power BI, Tableau, ArcGIS, Claude, Gemini, Github Copilot

CERTIFICATIONS

Microsoft Certified Power BI Data Analyst Associate, Microsoft

Aug 2025

PUBLICATIONS

Propagator Rooting Method Direction of Arrival Estimation Based on Real Data, MILCOM 2021

Capon Root-MUSIC-like Direction of Arrival Estimation Based on Real Data, VTC 2021